

Möbius

The BN Flow and the Inverse of Zeta

Diego Rincón
Independent

Abstract

The Beurling–Nyman flow approximates $1/\zeta(s)$ on the critical line [1]. The natural series for $1/\zeta(s)$ is the Möbius series: $1/\zeta(s) = \sum_{n=1}^{\infty} \mu(n)n^{-s}$ for $\operatorname{Re}(s) > 1$. The Möbius function μ encodes the multiplicative structure of the integers. The Möbius randomness law: $\mu(n)$ behaves like a random ± 1 sequence (Chowla’s conjecture, open). We connect the BN flow coefficients to Möbius partial sums, show that RH is equivalent to $\sum_{n \leq x} \mu(n) = O(x^{1/2+\epsilon})$, and explain why the BN greedy algorithm may converge faster than the natural Möbius series while being harder to prove converges.

1 The Möbius Function

Definition 1 (Möbius function). For $n \geq 1$:

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 p_2 \cdots p_k \text{ (distinct primes),} \\ 0 & \text{if } n \text{ is divisible by a perfect square } > 1. \end{cases}$$

The Möbius function is the arithmetic inverse of the constant function 1: $\sum_{d|n} \mu(d) = [n = 1]$ (Kronecker delta).

Theorem 2 (Möbius series for $1/\zeta$). For $\operatorname{Re}(s) > 1$:

$$\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}.$$

*This follows from $\zeta(s) = \sum_n n^{-s}$ and the convolution inverse $\mu * 1 = \delta$.*

Remark 3 (Möbius and RH). The prime number theorem is equivalent to $\sum_{n \leq x} \mu(n) = o(x)$. The Riemann Hypothesis is equivalent to $\sum_{n \leq x} \mu(n) = O(x^{1/2+\epsilon})$ for every $\epsilon > 0$. The Möbius randomness law (Chowla’s conjecture [5]): μ is asymptotically “random,” uncorrelated with any fixed sequence of ± 1 ’s. This remains open and is stronger than RH.

2 Two Approximations to $1/\zeta$

Proposition 4 (Two Dirichlet polynomial approximations). *There are two natural Dirichlet polynomial approximations to $1/\zeta(s)$ on the critical line:*

1. *The natural truncation: $M_N(s) = \sum_{n=1}^N \mu(n)n^{-s}$.*
2. *The BN optimal approximation: $P_N(s) = \sum_{k=1}^N c_k^*(k^{-s} - k^{-1})$, where c_k^* are the optimal BN coefficients.*

Both approximate $1/\zeta(s)$ in L^2 on the critical line. The BN approximation is optimal by construction (minimizes d_N^2). The Möbius truncation is natural (uses the exact series coefficients).

Proposition 5 (Mertens and BN entropy). *The Mertens function $M(x) = \sum_{n \leq x} \mu(n)$ controls the partial sum error:*

$$M_N(1/2 + it) - \frac{1}{\zeta(1/2 + it)} = - \sum_{n > N} \frac{\mu(n)}{n^{1/2+it}} = - \frac{1}{2\pi i} \int_{(2)} \frac{x^s}{s} \cdot \frac{1}{\zeta(s)} ds \Big|_{x=N} + \dots$$

The error is governed by $M(N)$. Under RH: $M(N) = O(N^{1/2+\epsilon})$, which gives convergence of $M_N(s)$ to $1/\zeta(s)$ in L^2 on $\text{Re}(s) = 1/2$.

Theorem 6 (BN entropy vs. Möbius error). *The BN entropy $d_N^2 = \|1_{(0,1)} - f_N^*\|^2$ and the Möbius error $e_N^2 = \|1_{(0,1)} - g_N^*\|^2$ where g_N^* = the function whose Mellin transform approximates $1/\zeta$ via M_N satisfy:*

$$d_N^2 \leq e_N^2$$

since the BN flow minimizes over all Dirichlet polynomials in the BN basis, while M_N uses a specific choice.

Consequence: if the Möbius approximation converges ($e_N^2 \rightarrow 0$, equivalent to RH via Mertens), then the BN approximation also converges ($d_N^2 \rightarrow 0$, equivalent to RH). The BN flow is at least as good as the natural series.

Remark 7 (Why BN is harder to analyze). Despite being optimal, the BN flow is harder to analyze than the Möbius series because:

1. The Möbius series has explicit coefficients $\mu(n)$ with known arithmetic properties. The BN coefficients c_k^* are defined implicitly via the Gram matrix.
2. The Möbius series can be studied via multiplicative functions, sieve theory, and analytic number theory. The BN coefficients require spectral theory of the Gram matrix G_N .
3. The Möbius randomness law has been extensively studied. The BN coefficient statistics are less understood.

The BN flow trades explicit structure for optimality.

3 The Möbius Randomness Law

Conjecture 8 (Chowla, 1965). *For any fixed $1 \leq a_1 < a_2 < \dots < a_k$ and any $\epsilon_1, \dots, \epsilon_k \in \{0, 1\}$ with $\sum \epsilon_i \geq 1$:*

$$\sum_{n \leq x} \mu(n + a_1)^{\epsilon_1} \dots \mu(n + a_k)^{\epsilon_k} = o(x).$$

The Möbius function is asymptotically uncorrelated with itself at different shifts.

Remark 9 (Sarnak's Möbius disjointness). Sarnak [6] conjectured that $\mu(n)$ is disjoint from all zero-entropy systems: for any zero-entropy topological dynamical system (X, T) and any continuous $f : X \rightarrow \mathbb{C}$: $\sum_{n \leq x} \mu(n) f(T^n x) = o(x)$. This connects Möbius to ergodic theory. The Sarnak conjecture is a weakening of Chowla's conjecture. Both remain open.

Theorem 10 (Möbius randomness and angle divergence). *If the Möbius randomness law holds in the following quantitative form:*

$$\left| \sum_{n \leq N} \mu(n) e^{it \log n} \right| = O(N^{1/2} \log^\alpha N) \quad \text{for all } t \in \mathbb{R} \text{ and some } \alpha < \infty,$$

then the BN angle sequence satisfies $\sum \cos^2 \theta_N = \infty$, and therefore RH holds.

The proof sketch: the inner product $\langle r_{N-1}, \rho_{1/N} \rangle$ involves $\int (N^{-1/2-it} - N^{-1}) \overline{\zeta(1/2 + it)} \cdot r_{N-1}(t) dt$ where $r_{N-1}(t)$ is the Mellin transform of the residual. The Möbius quantitative bound controls $1/\zeta(1/2 + it)$ via $M_N(1/2 + it) \approx 1/\zeta(1/2 + it)$ and gives a lower bound on the alignment.

Remark 11 (What this gives and doesn't give). Theorem 10 would prove RH if the quantitative Möbius randomness is established independently. But the quantitative Möbius randomness ($M(x) = O(x^{1/2+\epsilon})$) IS equivalent to RH. The circle closes: proving Möbius randomness requires RH, and proving BN angle divergence requires Möbius randomness.

These are different reformulations of the same thing. Each reformulation adds a different tool: Möbius: arithmetic and multiplicativity. BN flow: analytic function theory in H^2 . Angle: geometric/spectral condition. None provides a new proof; all provide new names for the same wall.

4 The Möbius Orthogonality

Proposition 12 (Möbius orthogonality to the BN residual). *The residual $r_{N-1} = 1_{(0,1)} - f_{N-1}^*$ satisfies $r_{N-1} \perp \mathcal{V}_{N-1}$, i.e., $\langle r_{N-1}, \rho_{1/k} \rangle = 0$ for $k = 1, \dots, N-1$.*

The Möbius function gives an explicit function in $\mathcal{V}_\infty^\perp$ under $\neg RH$: if $\zeta(\rho_0) = 0$ with $\text{Re}(\rho_0) > 1/2$, then the function $x \mapsto x^{-\rho_0} 1_{(0,1)}(x)$ is orthogonal to $\mathcal{V}_\infty$.

Proof: $\langle x^{-\rho_0} 1_{(0,1)}, \rho_{1/k} \rangle = \int_0^1 x^{-\rho_0} \rho_{1/k}(x) dx = \hat{\rho}_{1/k}(\rho_0) \cdot (1 - \rho_0) = (k^{-\rho_0} - k^{-1}) \zeta(\rho_0) / \rho_0 \cdot (1 - \rho_0) = 0$ since $\zeta(\rho_0) = 0$.

Under RH, no such ρ_0 exists, so $(\mathcal{V}_\infty)^\perp \cap \text{span}\{1_{(0,1)}\} = \{0\}$, and $d_N \rightarrow 0$.

Theorem 13 (RH as orthogonality completeness). *The following are equivalent:*

1. *RH.*
2. $d_N \rightarrow 0$ (*BN criterion*).
3. $\sum \cos^2 \theta_N = \infty$ (*angle criterion* [2]).
4. *The BN greedy algorithm approximates $1/\zeta$ in L^2 ([1]).*
5. *The function $1_{(0,1)}$ has no component orthogonal to $\mathcal{V}_\infty$.*
6. *Every function in $\mathcal{V}_\infty^\perp$ is orthogonal to $1_{(0,1)}$.*
7. *There is no zero ρ_0 with $\operatorname{Re}(\rho_0) > 1/2$ (no function of the form $x^{-\rho_0} 1_{(0,1)}$ is in $\mathcal{V}_\infty^\perp$).*

The Möbius function provides the arithmetic witness for items (5)–(7): μ is the Möbius inverse of 1, and its partial sums $M_N(s)$ approximate $1/\zeta(s)$ at the same rate at which RH holds.

Remark 14 (The Möbius function is the key). The Möbius function $\mu(n)$ is:

- The arithmetic inverse of 1 under Dirichlet convolution.
- The coefficient of the series $1/\zeta(s) = \sum \mu(n)n^{-s}$.
- The function whose partial sums control the error in $d_N^2 \rightarrow 0$.
- The oracle for primality: $\mu(n) = 0$ iff n is not squarefree.

Its randomness is RH. Its exact cancellations are the primes. The BN flow approximates $1/\zeta$ using the specific basis $\{\rho_{1/k}\}$, which are not the monomials n^{-s} but have Mellin transforms proportional to $\zeta(s)k^{-s}$. The BN basis “sees” ζ at every step. The Möbius basis ignores ζ and uses only arithmetic. The two approaches are dual: BN looks from the analytic side (Mellin, ζ , Hardy spaces), Möbius looks from the arithmetic side (primes, divisors, sieve). RH is the statement that these two sides converge to the same approximation. This is the fold [4]: arithmetic and spectrum, held inside the same equation.

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